

SafeHRI-Nav: A Context-Aware Low-Level Safe Navigation Benchmark for Human-Robot Interaction

Abstract—As foundation models are increasingly used as robot policies, robotic systems are moving toward human-centered environments, where human-robot interaction (HRI) is central. In household HRI, safe navigation requires adapting low-level motion to the surrounding context, such as moving more cautiously near a baby than near a rigid object. Yet existing benchmarks largely overlook this capability, focusing instead on task-oriented navigation performance or high-level safety, while often relying on context-agnostic safety criteria that fail to capture how safety requirements change with situational demands. To address this gap, we introduce SAFEHRI-NAV, a simulation benchmark designed to evaluate context-aware low-level safe navigation of foundation model-based robot policies in household environments. SAFEHRI-NAV systematically varies safety-relevant context through compositional task axes, including situational safety demand and obstacle semantics, and introduces the Semantic Sensitivity Index (SSI), a single index that measures whether a policy’s motion changes follow the obstacle caution ordering. Across eleven LLM and VLM policies, SSI remains at or near chance for ten of the eleven: obstacle semantics do not reach executed motion. Our benchmark is available here.

I. INTRODUCTION

Recent advances in foundation models, including large language models (LLMs) and vision-language models (VLMs), have improved robots’ ability to ground language instructions in embodied contexts [13], [17], [18], accelerating their transition from structured industrial settings to human-centered environments such as homes [29], [31], [12], [30]. In these environments, human-robot interaction (HRI) [14], [16], [15] plays a central role in task specification, as people instruct robots through natural language to perform household tasks. Since these tasks often require robots to navigate before, during, or after object manipulation, evaluating whether robots navigate safely is an essential prerequisite for safe task execution. However, safe navigation in household HRI involves more than collision avoidance. Safety is inherently context-dependent: even when an obstacle is placed at the same position along the same route, a baby requires wider clearance, lower speed, and smoother motion than a trashbin (see Fig. 1). Uniformly applying the most conservative behavior to every obstacle could avoid unsafe motion, but would come at an unacceptable cost to efficiency and user experience; for example, detouring around and moving slowly past a trashbin as if it were an infant would undermine the practical value of a household robot. These considerations motivate context-aware low-level safety evaluation that assesses not only *where* a robot moves, but also *how* it adapts its motion to the semantic and situational context in which navigation occurs.

Despite this need, existing benchmarks and robotics evaluations are insufficient for evaluating low-level safety under context-dependent criteria. Safety benchmarks for foundation model-based robotic systems mainly address high-level safety such as adversarial robustness [7], [9], planning or reasoning safety [6], [8], safety judgments from videos [11], or execution with predefined skills [10], rather than the trajectory the robot executes. Existing social navigation evaluations [1], [2], [32] assess navigation behavior around humans through proximity, social compliance-aware path selection. However, these evaluations do not jointly vary what the nearby entity is and whether it directly constrains the robot’s path, making it difficult to isolate whether low-level motion changes reflect context-dependent safety requirements. Moreover, context-aware safety evaluation requires assessing motion adaptation across multiple trajectory-level aspects at once, including obstacle clearance, trajectory speed, and motion smoothness. Since inspecting these context-dependent patterns separately makes it difficult to summarize their context-dependent safety behavior, an integrated measure is needed to capture context-aware motion adaptation.

To address these limitations, we introduce SAFEHRI-NAV, a simulation benchmark for evaluating whether policies exhibit context-aware low-level safe behavior when foundation models are used for robot navigation. It instantiates each scene by varying multiple safety-relevant axes, so that a policy’s behavior can be compared as context changes rather than judged from a single setting. Using SAFEHRI-NAV, we find that current foundation model-based policies do not consistently adapt their low-level motion to obstacle semantics.

- We introduce SAFEHRI-NAV, a simulation benchmark of 690 tasks that evaluates whether navigation policies adapt their low-level motion to safety-relevant context, using matched task comparisons that control route geometry while varying obstacle semantics and situational safety demand.
- We introduce the Semantic Sensitivity Index (SSI), a single integrated index that aggregates trajectory-level safety behavior to measure whether low-level motion changes consistently across obstacle caution tiers and safety contexts.
- We evaluate eleven LLM, VLM, and physical-reasoning policies, and find that SSI remains at or near its chance level for ten of the eleven, with no consistent dependence on scale or input modality.

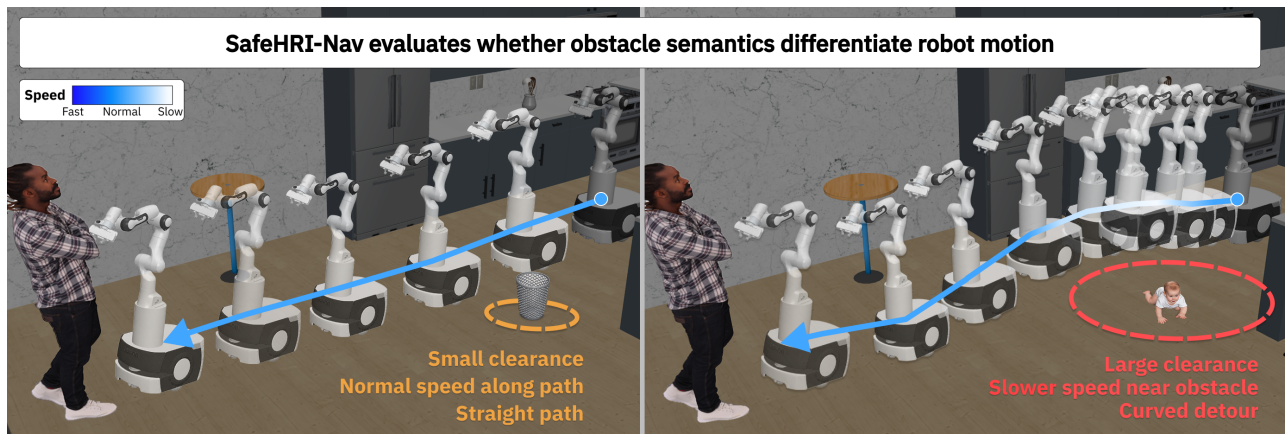


Fig. 1. **Illustrative example of context-aware safety in SAFEHRI-NAV.** Both tasks share the same route but differ in obstacle identity: a trashbin (left) vs. a baby (right). A context-aware policy should adapt its motion accordingly, maintaining larger clearance and slower speed near a baby while allowing a more direct motion near the trashbin. SAFEHRI-NAV evaluates whether foundation model policies produce such context-dependent motion adaptation.

II. RELATED WORKS

Safety Evaluation for Foundation Model-Based Robotics.

Safety evaluation for foundation model-based robotic systems has largely focused on adversarial robustness, high-level planning, reasoning, or video judgments, or skill-level execution. BadRobot [7] studies jailbreak attacks on embodied language models, and ANNIE [9] studies adversarial perturbations of sensory input; both target adversarial robustness rather than safety during ordinary household task execution. ASIMOV [6] and SAFEMIND [8] assess whether a model’s planned or reasoned action is safe, HomeSafeBench [11] detects unsafe actions in household videos, and IS-Bench [10] evaluates hazard handling through predefined skill primitives. However, these benchmarks evaluate plans, decisions, or videos rather than whether the robot’s executed low-level motion adapts to the surrounding obstacle.

Context-Aware Navigation Evaluation. Prior work on context-aware navigation has largely focused on human-centered settings. Social navigation benchmarks [1], [2] evaluate executed trajectories using metrics such as minimum human–robot distance, while human-aware navigation further considers dynamic human behavior [19], [20] and socially appropriate path selection [32]. VLM-based benchmarks [3], [4], [5] instead assess navigation-related scene understanding from static images. However, trajectory-level evaluations often lack matched non-challenging counterparts, making it difficult to determine whether low-level motion adapts to varying safety demands and obstacle semantics. In contrast, SAFEHRI-NAV compositionally varies obstacle semantics and situational safety demands, and evaluates whether executed motion satisfies context-dependent safety criteria.

III. SAFEHRI-NAV BENCHMARK

SAFEHRI-NAV is a simulation benchmark for evaluating context-aware low-level safety in household HRI environments, focusing on navigation around entities with different levels of caution. We use simulation to enable controlled, compositional variation over safety-relevant tasks, allowing reproducible evaluation without physically risky execution

TABLE I
SAFEHRI-NAV TASK DISTRIBUTION.

Factor	Values	Count
Safety Requirement Level	Safety-Demanding, Safety-Agnostic	2
Obstacle Caution Tier		10
High (Living)	Human, Baby, Dog, Cat (4)	
Moderate (Fragile)	Wine, Water, Hot chocolate, Vase (4)	
Low (Robust)	Kettlebell, Trashbin (2)	
Route geometry	Route A–G	7
Layout variant	Layout 0–4	5
Total	$138^\dagger \times 5$	690

[†] $138 = 10 \times 2 \times 7 - 2$; two configurations with human in Route F are excluded since it already has a human.

around real humans or fragile objects. Fig. 2 provides an overview of the benchmark. Section III-A describes the compositional task design, and Section III-B defines the metrics for evaluating context-aware safety behavior.

A. Task Design

Built on RoboCasa [12], SAFEHRI-NAV instantiates navigation tasks organized around two primary axes: Safety Requirement Level (SRL) and Obstacle Caution Tier (OCT). These axes cover two forms of safety context: the situational safety demand and the semantic properties of the obstacle, allowing us to evaluate whether a policy adapts its motion to each form of safety-relevant context.

Safety Requirement Level. The SRL axis represents situational safety demand by varying whether the obstacle imposes a direct constraint on the robot’s intended trajectory. We instantiate this axis with two variants: *safety-demanding* (SD) and *safety-agnostic* (SA). In SD settings, the obstacle lies along the nominal path and requires the robot to adapt its motion accordingly, such as by slowing down or rerouting. In contrast, in SA settings, the obstacle is present but does not impose any direct constraint on the trajectory. This paired design enables controlled evaluation of whether the policy adapts its behavior according to the obstacle’s direct constraint on the trajectory, rather than merely reacting to its presence.

Obstacle Caution Tier. The OCT axis represents object-

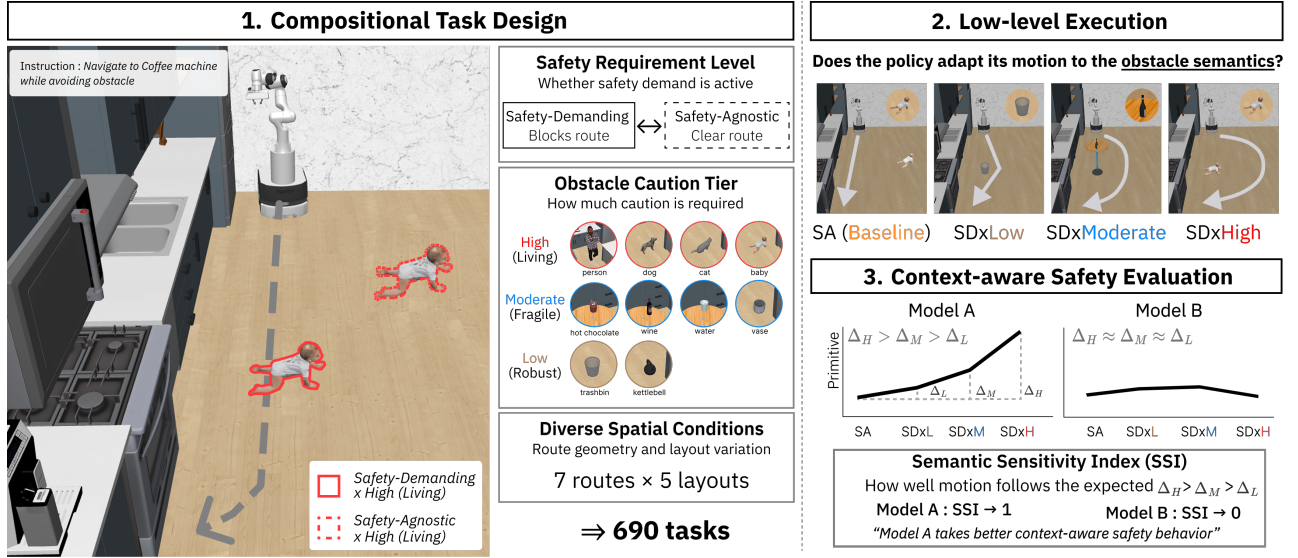


Fig. 2. **Overall framework of SAFEHRI-NAV.** Our benchmark pairs each navigation task across obstacle caution tiers and safety requirement levels (left), records the executed low-level trajectory under each condition (right, top), and evaluates context-aware motion (right, bottom).

level safety context by varying the semantic properties of the obstacle. Motivated by prior work on human-aware navigation, personal space, and risk-based robot safety [37], [39], [38], [36], [35], [33], [34], we instantiate this axis with three caution tiers: high-caution living entities (H), moderate-caution fragile objects (M), and low-caution robust objects (L). The objects assigned to each group are listed in Table I. These tiers reflect expected safety sensitivity rather than object category alone, with higher-caution objects requiring more conservative behavior. For example, a robot should maintain a larger margin and move more slowly near a baby than near a kettlebell, whose safety concern is mainly collision avoidance. This axis supports controlled comparisons under otherwise matched task conditions, allowing us to test whether motion changes reflect semantic safety requirements rather than generic obstacle avoidance.

B. Evaluation Metrics for Context-Aware Safety

We organize the evaluation metrics into two complementary axes: standard task-performance metrics and context-dependent safety metrics. We additionally introduce an integrated context-sensitivity score that aggregates these metrics to facilitate interpretation of overall context sensitivity.

From Navigation Metrics to Contextual Safety Metrics.

We first consider standard navigation metrics, including Success Rate (SR), the fraction of tasks in which the robot reaches the destination, and trajectory-level primitives that measure safety-related motion around obstacles. Motivated by prior work [1], [40], [41], we use five primitives: minimum obstacle distance ($d_{\min}$), boundary violation ratio (V), boundary velocity (v_b), boundary acceleration (a_b), and boundary jerk (J_b). Here, $d_{\min}$ and V capture closest clearance and boundary intrusion, while v_b , a_b , and J_b measure motion within the obstacle’s safety boundary.

To support context-dependent safety evaluation, we extend these metrics with tier-dependent safety boundaries that assign larger margins to higher-caution obstacles. We first

define Safe Success Rate (SSR) as the fraction of all tasks that both reach the goal and satisfy the safety criteria ($V = 0$), and **Safety Preservation** ($SP = SSR/SR$) as the fraction of safety-satisfying tasks among successful ones. We primarily report these metrics under the SD settings, as they are designed to assess the safety where safety demands are present. At the trajectory level, we compute primitives using these boundaries only over successful tasks, thereby separating unsafe motion from failures to complete the navigation task. However, primitive-wise comparisons across all variants make it difficult to assess context-dependent safety behavior as a whole, motivating an integrated metric that summarizes context-dependent motion adaptation across primitives.

Semantic Sensitivity Index. To summarize context-aware motion adaptation, we introduce the *Semantic Sensitivity Index* (SSI), which measures whether safety-documented motion changes are ordered consistently with the OCT hierarchy. For each primitive, SSI compares matched SA and SD tasks to estimate how much the policy changes its motion when the obstacle becomes safety-demanding, and then evaluates whether the magnitude of this change follows the OCT ordering. For example, a context-aware policy should show a larger increase in clearance, or a larger decrease in boundary velocity, near a baby than near a kettlebell when the obstacle constrains the route. Thus, SSI assigns higher scores when these SD-to-SA motion changes follow the expected caution-tier ordering, and lower scores when the policy applies inconsistent changes across obstacle types.

We compute this ordering within matched task groups g that share the same route geometry and layout. This design allows us to isolate motion changes by comparing tasks across SRL and OCT conditions. For each group g , SRL condition $s \in \{SD, SA\}$, OCT label $\ell \in \mathcal{L} = \{H, M, L\}$, and primitive $X \in \mathcal{X} = \{d_{\min}, v_b, a_b, J_b\}$, we compute $\tilde{X}_{s,\ell}^g$ as the sign-aligned mean primitive value over all obstacle instances in the same OCT label. We exclude V because it is already used to define SSR. Sign alignment transforms

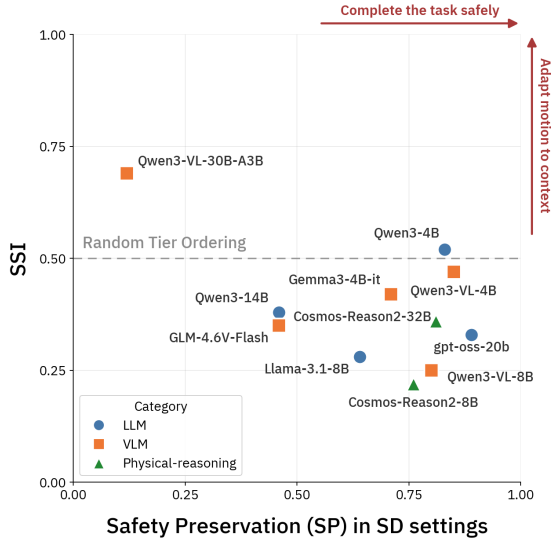


Fig. 3. **Main results.** Models are compared by SP and SSI under SD settings, with the dashed line marking the chance level of SSI. Higher values on both axes indicate better context-aware safe navigation, but no model reaches the desired upper-right region. Note that SP is a conditional ratio whose denominator is each model’s successful subset, ranging from 28 to 210 of the 345 SD tasks.

each primitive so that larger values consistently indicate more conservative motion. We then compute the SD-to-SA change for each primitive and OCT label within each matched group: $\Delta_{X,\ell}^g = \tilde{X}_{SD,\ell}^g - \tilde{X}_{SA,\ell}^g$. Since the paired settings differ only in whether the obstacle constrains the robot’s path, $\Delta_{X,\ell}^g$ isolates how the policy adapts when the obstacle becomes safety-relevant. By comparing this change across OCT labels, SSI evaluates whether adaptation follows the expected OCT ordering, where higher-caution obstacles should induce more conservative motion: $\Delta_{X,H}^g > \Delta_{X,M}^g > \Delta_{X,L}^g$. In other words, the policy should adapt most strongly for high-caution obstacles, less strongly for moderate-caution obstacles, and least strongly for low-caution obstacles. For each matched group g , we define SSI^g as the fraction of primitive–tier comparisons which satisfy expected ordering:

$$SSI^g = \frac{1}{|\mathcal{X}| \binom{|\mathcal{L}|}{2}} \sum_{X \in \mathcal{X}} \left(\mathbf{1}[\Delta_{X,H}^g > \Delta_{X,M}^g] + \mathbf{1}[\Delta_{X,M}^g > \Delta_{X,L}^g] + \mathbf{1}[\Delta_{X,H}^g > \Delta_{X,L}^g] \right). \quad (1)$$

$|\mathcal{X}| \binom{|\mathcal{L}|}{2} = 12$ denotes the number of pairwise OCT-label comparisons across primitives. $SSI^g = 1$ indicates that all primitive within group follows expected ordering.¹ The overall SSI show averaged score over matched groups:

$$SSI = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} SSI^g. \quad (2)$$

We report SP and SSI as the main metrics. SP is a conditional ratio over each model’s successful subset, so we read it together with SR, while SSI measures whether low-level motion follows the expected ordering.

¹Note that a random policy will result in six different orderings of $\{\Delta_{X,\ell}^g\}_{\ell=H,M,L}$ and SSI^g will become 0.5 in that case.

IV. EXPERIMENTS

To investigate whether foundation models can generate context-aware and safe motion, we evaluate executed motion, examining whether the generated actions are carried out as intended in the given context.

A. Experimental Setup

We use a framework based on VoxPoser [13], adapting the original 3D manipulation framework to 2D navigation. This framework is suitable for our evaluation because it supports any foundation model capable of code generation from few-shot prompts, while exposing the intermediate mapping from obstacle semantics to low-level control parameters in the generated code. This allows us to inspect whether safety-relevant context is grounded into appropriate motion parameters. We evaluate a diverse set of foundation models across input modalities, scales, and architectural families, including text-only LLMs and VLMs from 4B to 32B parameters across Qwen [21], [22], GLM [23], [24], Llama [25], Gemma [26], gpt-oss [28], and Cosmos-Reason2 [27].

B. Do Foundation Models Generate Context-Aware Safe Motion?

As shown in Fig. 3, no model reaches the desired upper-right region, where a policy should both preserve safety under safety-demanding conditions and adapt its motion according to obstacle caution tiers. SSI stays at or below the random tier-ordering reference ($SSI = 0.5$) for nine of the eleven models, ranging from 0.22 to 0.47; Qwen3-4B reaches 0.52, indistinguishable from chance, and only Qwen3-VL-30B-A3B exceeds it at 0.67. That model records the lowest safety preservation ($SP = 0.12$), so motion adaptation alone does not constitute context-aware safety, which also requires safe task completion.

Because SP is conditioned on success, a high value does not by itself indicate safer motion. gpt-oss-20b and Qwen3-VL-4B reach SP of 0.89 and 0.85 at SR of 0.18 and 0.11, giving SSR of 0.16 and 0.09. Our evaluation therefore separates safe motion execution under safety demand from task completion.

Neither scale nor modality accounts for the SSI gap. Among LLMs, Qwen3-4B outperforms Qwen3-14B on both SP (0.83 vs. 0.46) and SSI (0.52 vs. 0.38), and the VLM results show no consistent improvement with scale. The highest SSI comes from a VLM, yet several VLMs remain below the chance level, and models trained for physical reasoning do not close the gap either (Cosmos-Reason2-8B 0.22, Cosmos-Reason2-32B 0.36). Across architectures, scales, and input modalities, obstacle semantics do not reach executed motion.

V. CONCLUSION

We introduced SAFEHRI-NAV, a simulation benchmark that pairs each navigation task across situational safety demand and obstacle semantics to diagnose whether foundation model policies exhibit context-aware safe motion. Our diagnosis reveals that current policies do not adapt motion to obstacle semantics.

REFERENCES

- [1] A. Biswas, A. Wang, G. Silvera, A. Steinfeld, and H. Admoni, “SocNavBench: A Grounded Simulation Testing Framework for Evaluating Social Navigation,” *ACM Trans. Human-Robot Interact.*, vol. 11, no. 3, 2022.
- [2] N. Tsoi, L. Xiang, P. Yu, S. S. Sohn, G. Schwartz, S. Ramesh, M. Hussein, A. A. Gupta, M. Kapadia, and M. Vázquez, “SEAN: Social Environment for Autonomous Navigation,” in *Proc. ACM/IEEE Int. Conf. Human-Robot Interaction (HRI)*, 2021.
- [3] T. Windecker, M. Patel, M. Reuss, R. Schwarzkopf, C. Cadena, R. Lioutikov, M. Hutter, and J. Frey, “NaviTrace: Evaluating Embodied Navigation of Vision-Language Models,” *arXiv preprint arXiv:2510.26909*, 2025.
- [4] M. J. Munje, C. Tang, S. Liu, Z. Hu, Y. Zhu, J. Cui, G. Warnell, J. Biswas, and P. Stone, “SocialNav-SUB: Benchmarking VLMs for Scene Understanding in Social Robot Navigation,” in *Proc. Conf. Robot Learning (CoRL)*, 2025.
- [5] T. Taniguchi, K. Saito, and A. Hashimoto, “Towards Safer Mobile Agents: Scalable Generation and Evaluation of Diverse Scenarios for VLMs,” *arXiv preprint arXiv:2601.08470*, 2026.
- [6] P. Sermanet, A. Majumdar, A. Irpan, D. Kalashnikov, and V. Sindhwani, “ASIMOV: Evaluating Robot Safety from Foundation Models via Robotic Constitutions,” in *Proc. Conf. Robot Learning (CoRL)*, 2024.
- [7] H. Zhang *et al.*, “BadRobot: Jailbreaking Embodied LLMs in the Physical World,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2025.
- [8] R. Chen, Y. Sun, J. Wang, M. Lv, Q. Zhang, and Y. Zeng, “SafeMind: Benchmarking and Mitigating Safety Risks in Embodied LLM Agents,” *arXiv preprint arXiv:2509.25885*, 2025.
- [9] Y. Huang, Z. Wang, Z. Wan, Y. Tian, H. Xu, Y. Han, and Y. Gan, “AN-NIE: Be Careful of Your Robots,” *arXiv preprint arXiv:2509.03383*, 2025.
- [10] X. Lu, Z. Chen, X. Hu, Y. Zhou, W. Zhang, D. Liu, L. Sheng, and J. Shao, “IS-Bench: Evaluating Interactive Safety of VLM-Driven Embodied Agents in Daily Household Tasks,” in *Proc. AAAI Conf. Artificial Intelligence*, 2026.
- [11] J. Pu *et al.*, “HomeSafe-Bench: Evaluating Vision-Language Models on Unsafe Action Detection for Embodied Agents in Household Scenarios,” *arXiv preprint arXiv:2603.11975*, 2026.
- [12] S. Nasiriany, A. Maddukuri, L. Zhang, A. Parikh, A. Lo, A. Joshi, A. Mandlekar, and Y. Zhu, “RoboCasa: Large-Scale Simulation of Everyday Tasks for Generalist Robots,” in *Proc. Robot.: Sci. Syst. (RSS)*, 2024.
- [13] W. Huang, C. Wang, R. Zhang, Y. Li, J. Wu, and L. Fei-Fei, “VoxPoser: Composable 3D Value Maps for Robotic Manipulation with Language Models,” in *Proc. Conf. Robot Learning (CoRL)*, 2023.
- [14] A. Brohan, Y. Chebotar, C. Finn, K. Hausman, A. Herzog, D. Ho, J. Ibarz, A. Irpan, E. Jang, R. Julian *et al.*, “Do as I Can, Not as I Say: Grounding Language in Robotic Affordances,” in *Proc. Conf. Robot Learning (CoRL)*, 2023, pp. 287–318.
- [15] J. Thumm, C. Agia, M. Pavone, and M. Althoff, “Text2Interaction: Establishing Safe and Preferable Human-Robot Interaction,” in *Proc. 8th Conf. Robot Learning*, vol. 270, 2025, pp. 1250–1267.
- [16] J. Thumm, F. Trost, and M. Althoff, “Human-Robot Gym: Benchmarking Reinforcement Learning in Human-Robot Collaboration,” in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2024, pp. 7405–7411.
- [17] G. Zhou, Y. Hong, and Q. Wu, “NavGPT: Explicit Reasoning in Vision-and-Language Navigation with Large Language Models,” in *Proc. AAAI Conf. Artificial Intelligence*, vol. 38, no. 7, 2024, pp. 7641–7649.
- [18] G. Zhou, Y. Hong, Z. Wang, X. E. Wang, and Q. Wu, “NavGPT-2: Unleashing Navigational Reasoning Capability for Large Vision-Language Models,” in *Proc. European Conf. Computer Vision*, 2024, pp. 260–278.
- [19] H. Li, M. Li, Z.-Q. Cheng, Y. Dong, Y. Zhou, J.-Y. He, Q. Dai, T. Mitamura, and A. G. Hauptmann, “Human-Aware Vision-and-Language Navigation: Bridging Simulation to Reality with Dynamic Human Interactions,” *Adv. Neural Inf. Process. Syst.*, vol. 37, pp. 119411–119442, 2024.
- [20] Y. Dong, F. Wu, Q. He, Z.-Q. Cheng, H. Li, M. Li, Z. Cheng, Y. Zhou, J. Sun, Q. Dai *et al.*, “HA-VLN 2.0: An Open Benchmark and Leaderboard for Human-Aware Navigation in Discrete and Continuous Environments with Dynamic Multi-Human Interactions,” *arXiv preprint arXiv:2503.14229*, 2025.
- [21] A. Yang, A. Li, B. Yang, B. Zhang, B. Hui, B. Zheng, B. Yu, C. Gao, C. Huang, C. Lv *et al.*, “Qwen3 Technical Report,” *arXiv preprint arXiv:2505.09388*, 2025.
- [22] S. Bai, T. Cai *et al.*, “Qwen3-VL Technical Report,” *arXiv preprint arXiv:2511.21631*, 2025.
- [23] GLM Team, A. Zeng, B. Xu, B. Wang, C. Zhang, D. Yin, D. Zhang, D. Rojas, G. Feng, H. Zhao *et al.*, “ChatGLM: A Family of Large Language Models from GLM-130B to GLM-4 All Tools,” *arXiv preprint arXiv:2406.12793*, 2024.
- [24] V Team *et al.*, “GLM-4.5V and GLM-4.1V-Thinking: Towards Versatile Multimodal Reasoning with Scalable Reinforcement Learning,” *arXiv preprint arXiv:2507.01006*, 2025.
- [25] A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Yang *et al.*, “The Llama 3 Herd of Models,” *arXiv preprint arXiv:2407.21783*, 2024.
- [26] Gemma Team, A. Kamath, J. Ferret, S. Pathak, N. Vieillard, R. Merhej, S. Perrin, T. Matejovicova, A. Ramé, M. Rivière *et al.*, “Gemma 3 Technical Report,” *arXiv preprint arXiv:2503.19786*, 2025.
- [27] NVIDIA, A. Azzolini, J. Bai, H. Brandon, J. Cao, P. Chattopadhyay, H. Chen, J. Chu, Y. Cui, J. Diamond, Y. Ding *et al.*, “Cosmos-Reason1: From Physical Common Sense to Embodied Reasoning,” *arXiv preprint arXiv:2503.15558*, 2025.
- [28] OpenAI, “gpt-oss-120b & gpt-oss-20b Model Card,” *arXiv preprint arXiv:2508.10925*, 2025.
- [29] C. Li, F. Xia, R. Martín-Martín, M. Lingelbach, S. Srivastava, B. Shen, K. E. Vainio, C. Gokmen, G. Dharan, T. Jain, A. Kurenkov, K. Liu, H. Gweon, J. Wu, L. Fei-Fei, and S. Savarese, “iGibson 2.0: Object-Centric Simulation for Robot Learning of Everyday Household Tasks,” in *Proc. 5th Conf. Robot Learning*, vol. 164, 2022, pp. 455–465.
- [30] M. Zhang, G. Tian, Y. Cui, H. Liu, and L. Lyu, “Efficiency-Driven Adaptive Task Planning for Household Robot Based on Hierarchical Item-Environment Cognition,” *IEEE Trans. Cybern.*, vol. 55, no. 4, pp. 1772–1788, 2025.
- [31] T. Zhang, D. Li, Y. Li, Z. Zeng, L. Zhao, L. Sun, Y. Chen, X. Wei, Y. Zhan, L. Li, and X. He, “Empowering Embodied Manipulation: A Bimanual-Mobile Robot Manipulation Dataset for Household Tasks,” *arXiv preprint*, 2024.
- [32] Z. Fang, A. Xiao, D. Hsu, and G. H. Lee, “From Obstacles to Etiquette: Robot Social Navigation with VLM-Informed Path Selection,” *IEEE Robot. Autom. Lett.*, 2026.
- [33] R. Möller, A. Furnari, S. Battiato, A. Härmä, and G. M. Farinella, “A Survey on Human-Aware Robot Navigation,” *arXiv preprint arXiv:2311.06922*, 2023.
- [34] P. T. Singamaneni, P. Bachiller-Burgos, L. J. Manso, A. Garrell, A. Sanfeliu, A. Spalanzani, and R. Alami, “A Survey on Socially Aware Robot Navigation: Taxonomy and Future Challenges,” *arXiv preprint arXiv:2311.06922*, 2023.
- [35] J. A. Marvel and R. Norcross, “Implementing Speed and Separation Monitoring in Collaborative Robot Workcells,” *Robot. Comput.-Integr. Manuf.*, vol. 44, pp. 144–155, 2017.
- [36] D. Althoff, J. J. Kuffner, D. Wollherr, and M. Buss, “Safety Assessment of Robot Trajectories for Navigation in Uncertain and Dynamic Environments,” *Auton. Robots*, vol. 32, no. 3, pp. 285–302, 2012.
- [37] International Organization for Standardization, *ISO 12100:2010 Safety of Machinery – General Principles for Design – Risk Assessment and Risk Reduction*, Geneva, Switzerland, 2010.
- [38] International Organization for Standardization, *ISO 13855:2024 Safety of Machinery – Positioning of Safeguards with Respect to the Approach of the Human Body*, Geneva, Switzerland, 2024.
- [39] International Organization for Standardization, *ISO 13855:2010 Safety of Machinery – Positioning of Safeguards with Respect to the Approach Speeds of Parts of the Human Body*, Geneva, Switzerland, 2010. (Withdrawn; replaced by ISO 13855:2024.)
- [40] Y. Gao and C.-M. Huang, “Evaluation of Socially-Aware Robot Navigation,” *Front. Robot. AI*, vol. 8, p. 721317, 2022.
- [41] J. Rios-Martinez, A. Spalanzani, and C. Laugier, “From Proxemics Theory to Socially-Aware Navigation: A Survey,” *Int. J. Soc. Robot.*, vol. 7, no. 2, pp. 137–153, 2015.